<演習解説> 8の字運動ノードを作ってみよう

my_node.pyの変更

```
import rclpy
from rclpy.node import Node
from geometry_msgs.msg import Twist
from turtlesim.msg import Pose
import math

class EightFigureController(Node):
    def __init__(self):
        super().__init__('eight_figure_controller')

        self.linear_speed = 1.0
        self.angular_speed = 2.0
        self.current_angular_z = self.angular_speed

        self.last_theta = None
        self.cumulative_angle = 0.0

        self.publisher_ = self.create_publisher(Twist, '/turtle1/cmd_vel', 10)
        self.subscriber_ = self.create_subscription(Pose, '/turtle1/pose', self.pose_callback, 10)

        self.timer = self.create_timer(0.1, self.timer_callback)

    def pose_callback(self, msg):
        theta = msg.theta

        if self.last_theta is not None:
            # 差分を計算し、-π〜+πにラップ
            delta = theta - self.last_theta
            delta = (delta + math.pi) % (2 * math.pi) - math.pi  # 差分を[-π, π]に変換
            self.cumulative_angle += delta

            # 1回転（約2π）以上で方向反転
            if abs(self.cumulative_angle) > 2 * math.pi:
                self.current_angular_z *= -1
                self.cumulative_angle = 0.0
                self.get_logger().info('→ 方向反転')

        self.last_theta = theta

    def timer_callback(self):
        msg = Twist()
        msg.linear.x = self.linear_speed
        msg.angular.z = self.current_angular_z
        self.publisher_.publish(msg)

def main(args=None):
    rclpy.init(args=args)
    node = EightFigureController()
```

```
    try:
        rclpy.spin(node)
    except KeyboardInterrupt:
        pass
    finally:
        node.destroy_node()
        rclpy.shutdown()
```

パッケージのビルド

```
cd ~/colcon_ws
colcon build
source install/setup.bash
```

別ターミナルでTurtlesimを起動

```
ros2 run turtlesim turtlesim_node
```

元ターミナルで自作ノードを起動

```
ros2 run my_py_pkg my_node
```